

A Complemented-Kernel Partial Result for arXiv:2406.02330

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Source And Target

The source paper is L. Abadias, F. J. Gonzalez-Dona, and J. Oliva-Maza, *Universality arising from invertible weighted composition operators*, arXiv:2406.02330.

In Section 5 the authors pass from Hilbert spaces to the Banach spaces

$$\mathcal{B} = H^p \quad \text{or} \quad \mathcal{B} = \mathcal{A}_\sigma^p, \quad 1 \leq p < \infty, \quad \sigma > -1.$$

Let uC_ψ be an invertible weighted composition operator with hyperbolic symbol ψ , with attractive and repulsive fixed points a, b , and let

$$A^+ = \limsup_{z \rightarrow a} |u(z)|, \quad A^- = \liminf_{z \rightarrow a} |u(z)|, \quad B^+ = \limsup_{z \rightarrow b} |u(z)|, \quad B^- = \liminf_{z \rightarrow b} |u(z)|.$$

The parameter γ is $1/p$ for H^p and $(\sigma + 2)/p$ for $\mathcal{A}_\sigma^p$. Under

$$\frac{B^+}{(\psi'(b))^\gamma} < \frac{A^-}{(\psi'(a))^\gamma}$$

and

$$\frac{B^+}{(\psi'(b))^\gamma} < |\lambda| < \frac{A^-}{(\psi'(a))^\gamma},$$

the source asks whether $\ker(uC_\psi - \lambda I)$ is complemented and contains a subspace isomorphic to $\mathcal{B}$.

the following result:

Theorem 5.2. *Let uC_ψ be an invertible weighted composition operator with hyperbolic symbol ψ acting on $\mathcal{B}$ such that $\frac{B^+}{(\psi'(b))^\gamma} < \frac{A^-}{(\psi'(a))^\gamma}$. Then, for every $\frac{B^+}{(\psi'(b))^\gamma} < |\lambda| < \frac{A^-}{(\psi'(a))^\gamma}$, $\ker(uC_\psi - \lambda I)$ is infinite dimensional and $uC_\psi - \lambda I$ is surjective.*

Observe that this result is not sufficient to obtain the universality for the operators $uC_\psi - \lambda I$ acting on the Banach spaces $\mathcal{B}$. Indeed, by Theorem 5.1 it would be enough to answer the following question, that remains open, at least for the knowledge of the authors:

Question. Let uC_ψ be an invertible weighted composition operator with hyperbolic symbol acting on $\mathcal{B}$, where $\mathcal{B}$ denotes the Hardy space H^p or the weighted Bergman space $\mathcal{A}_\sigma^p$, where $1 \leq p < \infty$ and $\sigma > -1$. Assume that $\frac{B^+}{(\psi'(b))^\gamma} < \frac{A^-}{(\psi'(a))^\gamma}$ and let $\frac{B^+}{(\psi'(b))^\gamma} < |\lambda| < \frac{A^-}{(\psi'(a))^\gamma}$. Is $\ker(uC_\psi - \lambda I)$ a complemented subspace that contains a subspace which is isomorphic to $\mathcal{B}$?

We also observe that the analogous results of Proposition 3.2, Proposition 3.3 and Theorem 4.3 adapted to H^p and $\mathcal{A}_\sigma^p$ considerably improve previous results in the literature. Indeed, the spectrum of uC_ψ was given in [16, Theorem 4.9] provided that the function u belongs to the disc algebra (i.e. u has continuous extension to $\mathbb{D}$, and then we write $A := A^+ = A^-$ and $B := B^+ = B^-$) and the

Result

We prove the complementability half of the question.

Theorem 1. *Under the hypotheses above, the operator*

$$S_\lambda = \lambda I - uC_\psi : \mathcal{B} \rightarrow \mathcal{B}$$

has a bounded right inverse. Consequently

$$\ker(uC_\psi - \lambda I) = \ker S_\lambda$$

is a complemented subspace of $\mathcal{B}$.

Proof Intuition

The source already proves that S_λ is onto $\mathcal{B}$, but it does so by solving the equation separately on two range spaces $\mathcal{B}_{\mu,0}$ and $\mathcal{B}_{0,\nu}$, then using the algebraic decomposition $\mathcal{B} = \mathcal{B}_{\mu,0} + \mathcal{B}_{0,\nu}$. The small extra observation is that this decomposition can be chosen by a bounded linear formula. Therefore the two bounded inverses on the range spaces glue into a bounded right inverse on $\mathcal{B}$. A bounded right inverse immediately gives a projection onto the kernel.

Proof

For $\mu, \nu \geq 0$, set

$$\omega_{\mu,\nu}(z) = (a-z)^\mu(b-z)^\nu, \quad \mathcal{B}_{\mu,\nu} = \omega_{\mu,\nu}\mathcal{B}$$

with norm $\|f\|_{\mathcal{B}_{\mu,\nu}} = \|f/\omega_{\mu,\nu}\|_{\mathcal{B}}$. Multiplication by $\omega_{\mu,\nu}$ identifies $\mathcal{B}$ isometrically with $\mathcal{B}_{\mu,\nu}$, and the inclusion $\mathcal{B}_{\mu,\nu} \hookrightarrow \mathcal{B}$ is bounded.

Choose $\mu, \nu > 0$ as in the proof of Theorem 4.1 of the source, adapted in Section 5 to the Banach spaces: namely,

$$|\lambda| > A^+(\psi'(a))^{\mu-\gamma}, \quad |\lambda| < B^-(\psi'(b))^{\nu-\gamma}.$$

The source argument shows that S_λ is boundedly invertible on $\mathcal{B}_{\mu,0}$ and on $\mathcal{B}_{0,\nu}$. Denote these bounded inverses by

$$R_\mu : \mathcal{B}_{\mu,0} \rightarrow \mathcal{B}_{\mu,0}, \quad R_\nu : \mathcal{B}_{0,\nu} \rightarrow \mathcal{B}_{0,\nu}.$$

It remains only to keep track of the splitting. Pick integers m, n with $m \geq \mu$ and $n \geq \nu$. For $f \in \mathcal{B}$, the binomial identity gives

$$f(z) = \frac{1}{(a-b)^{m+n}} \sum_{j=0}^{m+n} \binom{m+n}{j} (-1)^{m+n-j} (a-z)^j (b-z)^{m+n-j} f(z).$$

Define

$$Pf = \frac{1}{(a-b)^{m+n}} \sum_{j=m}^{m+n} \binom{m+n}{j} (-1)^{m+n-j} (a-z)^j (b-z)^{m+n-j} f$$

and

$$Qf = \frac{1}{(a-b)^{m+n}} \sum_{j=0}^{m-1} \binom{m+n}{j} (-1)^{m+n-j} (a-z)^j (b-z)^{m+n-j} f.$$

Then $P, Q : \mathcal{B} \rightarrow \mathcal{B}$ are bounded linear maps, $P + Q = I_{\mathcal{B}}$, $\text{ran } P \subseteq \mathcal{B}_{\mu,0}$, and $\text{ran } Q \subseteq \mathcal{B}_{0,\nu}$.

Now set

$$R = R_{\mu}P + R_{\nu}Q : \mathcal{B} \rightarrow \mathcal{B}.$$

This operator is bounded, and

$$S_{\lambda}Rf = S_{\lambda}R_{\mu}Pf + S_{\lambda}R_{\nu}Qf = Pf + Qf = f$$

for every $f \in \mathcal{B}$. Hence R is a bounded right inverse for S_{λ} .

Finally,

$$\Pi = I_{\mathcal{B}} - RS_{\lambda}$$

is a bounded projection onto $\ker S_{\lambda}$: indeed

$$S_{\lambda}\Pi = S_{\lambda} - S_{\lambda}RS_{\lambda} = 0,$$

so $\text{ran } \Pi \subseteq \ker S_{\lambda}$, while $\Pi x = x$ for every $x \in \ker S_{\lambda}$. Thus $\ker(uC_{\psi} - \lambda I)$ is complemented. $\square$

Limitations And Novelty Check

This does not prove that the kernel contains a subspace isomorphic to $\mathcal{B}$, so it does not by itself prove Banach-space universality via the Banach Caradus criterion. It reduces the source question to the copy-of- $\mathcal{B}$ part.

The run's lightweight indexes had no exact prior packet for arXiv:2406.02330. An exact web search for the complemented-kernel phrasing found the source arXiv record and no separate answer.

References

References

- [1] L. Abadias, F. J. Gonzalez-Dona, and J. Oliva-Maza, *Universality arising from invertible weighted composition operators*, arXiv:2406.02330.
- [2] S. R. Caradus, *Universal operators and invariant subspaces*, Proc. Amer. Math. Soc. 23 (1969), 526–527.